Team 27 - Development Plan

with revisions made from lab 11.2

Team Members and Roles:

- Syaan Merchant - 400571626 – syaanmerchant - Interface & Medication System
- Shifa Zaman - 400563865 - ShifaZaman - Data Export & file management
- Mahnoor Naveed - 400565876 - mahnoornav - Medication Logic & ID Management

Title of Project: Medmate

High Level Description:

a) *What kind of app is it? What will it do, and how will the user use it?*
    - Revisions made: We simplified the notification system by removing the snooze option. Reminders now stay visible until the user chooses an action. File attachments are not supported yet, but users can add notes, and a new Wikipedia lookup feature was introduced. The interface was reorganized into clearer, separate sections to make navigation cleaner. Upon entering the app, users now immediately see a structured list showing each medication along with its dosage, time, and related details

    MedMate is a personal health management app designed to help users stay organized with their medications and healthcare appointments. It combines the core functions of medication reminders, appointment tracking, and medication information management into a single interface. Users can add medications with detailed information, such as name, dosage, and use, and create reminders based on these schedules. Notifications now remain visible until addressed, and users can add notes to their entries. Medical appointments can also be created and managed. MedMate includes a medication information database where users can store drug details. All data is stored locally for security and privacy. When the app is opened, users see a structured list of their medications, showing dosage, time, and related details

b) What is the context in which this app will be used (i.e., characteristics of the expected users and the environment in which they will use it)?
    - No specific revisions have been made in this portion; the target audience and the context in which the app will be used remain the same. It is important to note that the platform currently supports only a single account. While the app may reference a second-party user, they would be accessing it from the same account and viewing the reports. Implementing multi-account support has been identified as a potential next step for development

    MedMate is targeted towards those taking multiple medications consistently or managing health schedules for themselves or loved ones. Its most frequent users include students, working adults, and caregivers who require privacy, simplicity, and reliable reminders. For users with less technological experience, the interface will be made to be intuitive, complete with easy to use buttons, clear instructions, and help commands. Secondary users could include family caregivers or healthcare providers who may use the app indirectly by reviewing reports or appointment summaries. The usage of MedMate will take place predominantly in home settings on either laptops or computers. This allows patients to quickly access their medical notes and generate reports to be shared with their physicians. Since this app handles sensitive health data, MedMate's local-storage model ensures users maintain full control over their information without requiring cloud accounts or internet connectivity.

c) What constraints are there on your design (i.e., performance, memory, user interface, development time and resources, etc.)
    - Revisions Made: We updated the technical approach by specifying the reactive framework used, choosing GTK for the GUI to provide a responsive, cross-platform desktop application. Other revisions include simplifying the notification system by removing the snooze option, allowing users to add notes, and reorganizing the interface into clearer sections. Upon opening the app, users now see a structured list of medications showing dosage, time, and related details.

The MedMate design operates under technical and usability constraints that shape its maintainability. The app must be completely offline, storing all data locally in a database to ensure privacy and safeguard it against connectivity issues. The reminder system must be reliable, triggering notifications on time. Usability standards follow normal accessibility guidelines to support users with varied abilities, while the interface must remain minimal, responsive, and easy to navigate. From a development perspective, the app is built by a student team with limited knowledge of C GUI programming and within a restricted time frame. This prioritizes key features such as medication and appointment management, reminders, exporting, and basic reporting. The interface and GUI are implemented using GTK to provide a cross-platform desktop application.

Sections for Each Team Member

- Revisions made: We assigned each team member a specific code file to reduce errors from multiple people working in the same file. Responsibilities were clarified so that each person knows exactly what to work on, and redundant or repetitive sections were removed, giving everyone their own distinct section to manage

[Mahnoor Naveed]: Reminder and Sorting Module

Focus: Design and implement the core backend logic responsible for medication scheduling, ID assignment, time comparison, and recurrence validation within the MedMate application.

Main tasks: main code is to be done in the "core.c" file and should include.

- Functions for adding medications, assigning unique ID's, and sorting/organizing medication entries
- Implement due-time scanning across multiple dose windows, forming the backend of the reminder system
- Maintain data integrity and avoid errors by preventing overflow when the medication list reaches its capacity

[Shifa Zaman]: Data Export, File input/output, and Data Storage System

Focus: Implement the full file handling system for loading, saving, and exporting medication data. Ensure that all user entered medications persist between sessions and can be exported in a clean, readable format

Main tasks: main code is to be done in the "io.c" file and should include

- Develop all major file input/output operations used by medmate
- Converts the internal medication data into a CSV file readable by Excel and Google Sheets
- Ensures exported files are clean, structured, and user-friendly for reporting or printing

[Syaan Merchant] : Medication Database and GUI Module Developer

Focus: Build the main medication system and design the graphical user interface

Main tasks: main code is to the done in the "gui.c" file and should include

- Design and implement the full GUI layout for MedMate
- Connected the GUI with the backend so medication data appears correctly in the interface
- Come up with logic for wikipedia lookup and implementation

Increments

Increment 0 — Project Setup

Target Date: *November 15th, 2025*

Goals: Set up the basic environment, app structure, and database connection.

Deliverables:

- Syaan Merchant: Base UI layout (sidebar + Today Dashboard placeholder)
- Mahnoor Naveed: Simple test reminder system that prints console notifications.

- Shifa Zaman: Appointment input form stored in the local databases
- Shared: Repository setup

Increment 1

Target Date: *Lab 11.2 November 25th , 2025*
Goals: Have the basic medication reminder and appointment management working offline.
Deliverables:

- Mahnoor Naveed: work on the medication management logic, including adding and editing medications and handling ID updates.
- Shifa Zaman: work on the file handling and CSV export features for saving and loading medication data.
- Syaan Merchant: work on the main interface that connects the GUI to the medication system so users can interact with the app smoothly.

Demo Features:

- Add medications and appointments.
- Receive local reminders.
- View the Today Dashboard.
- Export adherence data as a CSV file.

Increment 2

Target Date: *November 30th, 2025*
Goals: Add finishing touches, improve usability, and finalize reports.
Deliverables:

- Mahnoor Naveed: ensure reminders persist after restart.
- Shifa Zaman: Notes system for each appointment; improved CSV formatting.
- Syaan Merchant: Enhanced Today Dashboard visuals and wikipedia implementation.
- Shared: input validation, error messages, and user testing.

Increment 3 — Final

Target Date: *December 3, 2025*
Goals: Polish, test, and package the finished app.
Deliverables (All):

- Fix bugs, ensure accessibility compliance, finalize styling.
- prepare demo and final documentation/submission.